

MERZIFON, Turkey

WMO: 170820

Lat: 40.836N

Lon: 35.526E

Elev: 1755

StdP: 13.79

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.9	(c) 19.0	(d) 10.0	(e) 9.8	(f) 16.8	(g) 13.8	(h) 11.8	(i) 20.3	(j) 22.6	(k) 37.8	(l) 20.0	(m) 37.4	(n) 2.1	(o) 10	(p) 0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 26.9	(c) 93.3	(d) 67.4	(e) 89.6	(f) 66.8	(g) 86.3	(h) 65.6	(i) 70.4	(j) 87.2	(k) 68.6	(l) 84.3	(m) 67.4	(n) 82.3	(o) 11.2	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	97.3	78.3	62.9	91.6	76.1	61.6	87.5	75.0	35.3	86.9	33.9	84.4	32.8	82.6	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.4	(b) 21.7	(c) 19.6	(e) 8.3	(f) 101.0	(g) 7.9	(h) 3.4	(i) 2.6	(j) 103.4	(k) -2.1	(l) 105.4	(m) -6.5	(n) 107.3	(o) -12.3	(p) 109.7		
			DB													
			WB													
				8.3	101.0	7.9	3.4	2.6	103.4	-2.1	105.4	-6.5	107.3	-12.3	109.7	
				7.1	73.3	7.8	2.2	1.5	74.9	-3.0	76.2	-7.4	77.5	-13.1	79.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	34.7	38.1	44.2	52.3	60.6	67.0	71.8	72.6	65.9	56.8	45.3	37.8		
	DBStd	14.74	8.53	7.83	7.58	6.95	6.30	4.95	4.51	4.51	5.73	6.75	6.87	7.19		
	HDD50	1654	475	338	210	56	4	0	0	0	0	17	172	382		
	HDD65	4680	939	754	644	384	162	34	3	2	58	265	592	843		
	CDD50	3116	1	4	31	124	333	509	676	701	476	227	30	4		
	CDD65	665	0	0	0	1	26	93	214	238	83	10	0	0		
	CDH74	7876	0	0	10	93	513	1165	2393	2498	1001	203	0	0		
	CDH80	2948	0	0	1	10	133	372	978	1046	359	49	0	0		
(14) Wind	WSAvg	6.5	4.2	5.4	6.6	6.6	6.4	7.8	9.6	10.0	7.8	5.7	3.5	3.8		
(15) Precipitation	PrecAvg	16.0	1.5	1.1	1.3	2.0	2.2	1.8	0.7	0.5	0.8	1.2	1.2	1.8		
	PrecMax	25.7	3.2	3.0	2.7	3.4	5.1	3.4	2.6	1.7	2.5	6.3	3.6	3.6		
	PrecMin	11.8	0.2	0.2	0.0	0.6	0.9	0.5	0.0	0.0	0.0	0.0	0.1	0.4		
	PrecStd	2.9	0.9	0.6	0.6	0.7	0.9	0.9	0.8	0.5	0.7	1.2	0.8	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	64.7	74.4	80.9	88.1	93.2	98.9	98.5	94.0	85.6	70.1	60.7		
		MCWB	48.0	49.6	55.4	59.7	63.6	66.3	68.6	67.9	64.8	62.0	55.0	50.7		
	2%	DB	53.5	60.0	68.0	76.8	83.9	87.7	93.3	93.1	88.2	79.2	66.1	55.6		
		MCWB	46.0	48.6	52.5	58.0	63.1	65.6	68.2	67.7	63.8	59.6	53.4	47.8		
	5%	DB	49.9	55.6	63.3	72.0	79.7	84.2	89.2	89.4	84.0	74.9	61.2	51.9		
		MCWB	43.8	46.4	50.0	56.1	61.5	65.1	67.1	67.6	63.4	58.5	51.2	45.8		
	10%	DB	46.3	51.6	58.9	67.8	75.6	80.7	85.6	86.0	79.9	69.8	57.3	48.4		
		MCWB	42.0	44.4	48.7	54.5	60.5	63.8	66.0	66.3	62.4	56.9	49.3	44.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	52.0	57.2	62.8	68.0	70.0	73.3	72.2	68.6	64.5	57.4	51.8		
		MCDB	56.6	62.0	71.1	76.7	83.8	87.0	90.5	89.9	86.0	79.5	67.5	59.6		
	2%	WB	46.4	48.9	53.3	59.3	64.7	67.7	70.6	70.3	66.6	61.3	54.2	48.5		
		MCDB	52.8	58.3	64.9	72.6	78.9	82.7	87.3	87.4	82.1	74.0	63.2	54.7		
	5%	WB	44.1	47.0	50.9	57.1	62.9	66.1	68.6	68.7	64.8	59.5	52.0	46.5		
		MCDB	49.0	55.3	61.5	70.0	76.4	80.8	83.9	84.4	78.9	70.6	60.3	51.4		
	10%	WB	41.9	44.5	48.8	55.0	61.2	64.6	67.1	67.5	63.2	58.1	49.7	44.4		
		MCDB	45.7	50.8	58.5	66.1	74.0	78.4	81.8	82.3	76.7	68.1	56.9	48.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	19.0	21.7	25.1	26.7	26.5	27.3	26.9	26.6	22.8	21.0	15.0		
		MCDBR	20.7	26.4	30.9	33.6	33.8	32.0	34.1	32.4	34.4	31.4	27.0	20.2		
		MCWBR	14.3	16.8	17.7	18.2	16.6	14.2	14.5	13.2	15.3	15.2	16.3	14.0		
	5% WB	MCDBR	19.5	24.9	27.8	30.1	30.3	29.0	29.3	28.4	29.6	26.5	24.5	18.1		
		MCWBR	13.8	16.2	17.0	17.6	16.0	13.8	13.1	12.6	14.3	13.6	15.4	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.351	0.394	0.435	0.410	0.386	0.393	0.397	0.372	0.369	0.332	0.319		
	taud		2.351	2.284	2.192	2.108	2.257	2.360	2.352	2.325	2.393	2.365	2.428	2.406		
	Ebn at Noon		262	270	269	266	275	282	279	274	274	260	257	253		
	Edh at Noon		27	34	42	49	43	39	39	39	34	31	25	24		
(44) All-Sky Solar Radiation	RadAvg		556	840	1158	1556	1885	2151	2212	1977	1528	1011	699	484		
	RadStd		56	124	110	166	172	167	111	103	104	90	78	59		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	N/A	N/A	N/A	N/A	+0.92	-175	-236	N/A	+94
(47) Regional (7 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	-148	-257	+191	+119

Nomenclature: See separate page